Environment Setup

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Outline

- Creating a VM on MacOS
- Getting started with Wireshark

Installing Multipass on MacOS

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- You can check the available Ubuntu images with the following command

```
multipass find
```

Creating a Ubuntu VM

- To create a Ubuntu 22.04 VM with 4 CPUs, 6GB RAM, and 50GB disk run the following command

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multipass launch 22.04 -n networking -c 4 -m 6G -d 50G
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- You can attach to the running VM with the following command

```
multipass shell networking
```

Creating a Ubuntu VM

- Make sure that the VM is up to date and install Ubuntu destop

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sudo apt-get upgrade -y
sudo apt-get install -y ubuntu-desktop xrdp
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- Create a new user account to login and add it to the sudo group to give him administrative access

```
sudo adduser <username>
sudo usermod -a -G sudo <username>
```

Connect to your VM

- Install an RDP client. You could use Microsoft Remote Desktop. You can install Microsoft Remote Desktop with the following command

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- Get the IP address of your VM and connect to it using the RDP client you installed. You can get the IP address of your VM with the following command

```
multipass list
```

Install Wireshark

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- On RPM-based distros, run the following commands

```
sudo dnf install wireshark
```

Install Wireshark

- (Optional) Add your user to the ***wireshark*** group to capture traffic with your user

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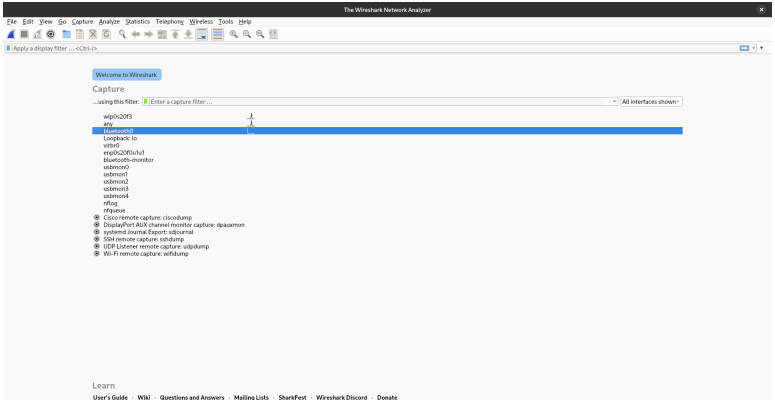
- Let's open Wireshark and play a bit with it



Wireshark filters

- There are two types of filters in Wireshark
  - The ***Capture filters*** are used to match packets into a stream of packets and to reduce the size of a packet capture.
  - The ***Display filters*** are used to hide/show packets from the packet list, color packets, etc.

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- You can find a guide about ***Display filters*** at ***https://wiki.wireshark.org/DisplayFilters***

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- Each primitive usually consist of one or more quantifiers followed by an id (a port number, an address, etc). The qualifiers are of three types
  - ***type***. The kind of thing the id refers to. (ie ***port***, ***portange***, ***host***, etc).
  - ***dir***. The transfer direction to or from the id. (ie ***src***, ***dst***, etc).
  - ***proto***. It restricts the match to a particular protocol. (ie ***udp***, ***tcp***, ***ether***, etc).

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- Filter expressions can be combined using logical operators (ie ***and***, ***or***, ***not***, etc)

Debugging capture filters

- The capture filters are compiled into BPF opcodes. You can find a guide about BPF at ***https://www.kernel.org/doc/Documentation/networking/filter.txt***

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- Lets debug together the filter: ip host 1.2.3.4
- Run the following command

```
dumpcap -f 'ip host 1.2.3.4' -d
```

Debugging capture filters

- The BPF opcodes for the filter are the following

```
(000) ldh [12]
(001) jeq #0x800 jt 2 jf 7
(002) ld [26]
(003) jeq #0x1020304 jt 6 jf 4
(004) ld [30]
(005) jeq #0x1020304 jt 6 jf 7
(006) ret #262144
(007) ret #0
```

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  - ip.addr == 192.168.1.0/24
  - ip.src == 10.43.54.65 or ip.dst == 10.43.54.65
  - tcp.flags.syn || tcp.flags.ack
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- You can build display filters interactively in the display filters bar of the Wireshark grafical interface